

Questions with Answer Keys

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Q1 - Single Correct

Let f be a real valued function such that $f(x) + 2f\left(\frac{2002}{x}\right) = 3x$, for all $x > 0$. The value of $f(2)$, is

- (1) 1000
- (2) 2000
- (3) 3000
- (4) 4000

Q2 - Single Correct

If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$, where lies in the interval, where $(a, b, c \in \text{are sides of } \Delta)$.

- (1) $\left[\frac{1}{2}, 2\right]$
- (2) $[-1, 2]$
- (3) $\left(\frac{1}{2}, 1\right]$
- (4) $\left[-1, \frac{1}{2}\right]$

Q3 - Single Correct

Given $f(x) = \sqrt{\frac{8}{1-x} + \frac{8}{1+x}}$ and $g(x) = \frac{4}{f(\sin x)} + \frac{4}{f(\cos x)}$. Then, $g(x)$ is

- (1) periodic with period 2
- (2) periodic with period $\frac{\pi}{2}$
- (3) periodic with period 2π
- (4) Non periodic

Q4 - Single Correct

Let f be a function satisfying $f(xy) = \frac{f(x)}{y}$ for all positive real numbers x and y . If $f(30) = 20$, then value of $f(40)$ is

- (1) 15
- (2) 20

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(3) 40

(4) 60

Q5 - Single Correct

Let $f(x) = \sin^2 x + \cos^4 x + 2$ and $g(x) = \cos(\cos x) + \cos(\sin x)$. Also, let period of $f(x)$ and $g(x)$ be T_1 and T_2 respectively, then

(1) $T_1 = 2T_2$

(2) $2T_1 = T_2$

(3) $T_1 = T_2$

(4) $T_1 = 4T_2$

Q6 - Single Correct

The minimum value of expression $|x - p| + |x - 15| + |x - p - 15|$ for x in the range $p \leq x \leq 15$, where $0 < p < 15$, is

(1) 10

(2) 15

(3) 30

(4) 0

Q7 - Single Correct

If $f(x, y) = [\max(x, y)]^{\min(x, y)}$ and $g(x, y) = \max(x, y) - \min(x, y)$, then $f[g(-1, -3/2), g(-4, -1.75)]$ equals

(1) -0.5

(2) 0.5

(3) 1

(4) 1.5

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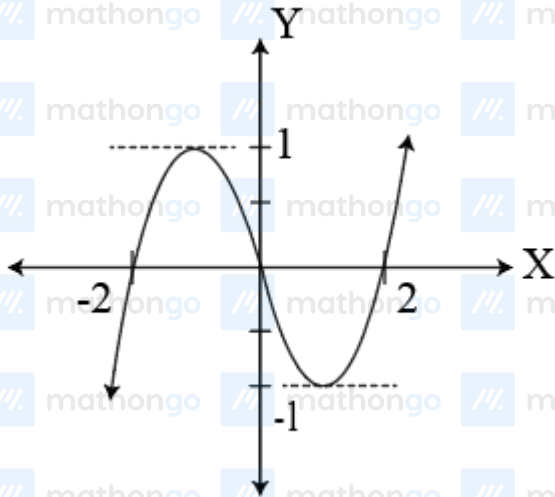
Q8 - Single Correct

If the solution set of $f(x) < 3$ is $(0, \infty)$ and the solution set of $f(x) > -2$ is $(-\infty, 5)$, then the solution set for $(f(x))^2 \geq f(x) + 6$ is

- (1) R
- (2) $(-\infty, 0]$
- (3) $[0, 5]$
- (4) $(-\infty, 0] \cup [5, \infty)$

Q9 - Single Correct

The graph of function $y = g(x)$ is shown, The number of solutions of equation $||g(x)| - 1| = \frac{1}{2}$ is



- (1) 4
- (2) 5
- (3) 6
- (4) 8

Q10 - Single Correct

$f(x) = x^2 + bx + c$ and $f(2+t) = f(2-t)$ for all real numbers t , then which of the following is true?

- (1) $f(1) < f(2) < f(4)$
- (2) $f(2) < f(1) < f(4)$

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(3) $f(2) < f(4) < f(1)$

(4) $f(4) < f(2) < f(1)$

Q11 - Single Correct

The solution of set of $[x]\{x\} = 1$, where $[x]$ and $\{x\}$ are greatest integer and fractional part of x , is

(1) $R^+ - (0, 1)$

(2) $R^+ - \{1\}$

(3) $\left\{m + \frac{1}{m} : m \in \mathbb{N}\right\}$

(4) $\left\{m + \frac{1}{m} : m \in \mathbb{N} - \{1\}\right\}$

Q12 - Single Correct

Let $f(x) = \frac{x}{1+x}$ and $g(x) = \frac{rx}{1-x}$. Let S be the set of all real numbers r such that $f[g(x)] = g[f(x)]$ for infinitely many real number x . Then, the number of elements in set S is

(1) 1

(2) 2

(3) 4

(4) 5

Q13 - Single Correct

A polynomial of 6th degree $f(x)$ satisfies $f(x) = f(2-x), \forall x \in \mathbb{R}$, if $f(x) = 0$ has 4 distinct and 2 equal roots, then the sum of roots of $f(x) = 0$, is

(1) 4

(2) 5

(3) 6

(4) 7

Q14 - Single Correct

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Let $f(x)$ be a polynomial satisfying $f(9x) = f'(3x) \cdot f''(3x)$, then which of the following is correct?

- (1) $f(x)$ is a polynomial of degree 3
- (2) $f(x)$ is a polynomial of degree 4
- (3) $f(x)$ is a polynomial of degree 5
- (4) Data insufficient

Q15 - Single Correct

Let $f : R \rightarrow R$ be a many one function and there exists a function $g : R \times R \rightarrow R$ such that

$$f(x + y) = g[f(x), y] \forall x, y \in R, \text{ then}$$

- (1) f is non-periodic
- (2) g is non-periodic
- (3) $f \cdot g$ may or may not be periodic
- (4) None of these

Q16 - Multiple Correct

Let $f : R - \{0, 1\} \rightarrow R$ satisfying $f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)}$, then

- (1) $y = f(x)$ crosses X -axis at $x = 1$
- (2) $f(2) + f(3) = 5$
- (3) $f(2) + f(3) = 4$
- (4) $f(2) + f(3) = 6$

Q17 - Multiple Correct

The function g defined by $g(x) = \sin \alpha + \cos \alpha - 1, \alpha = \sin^{-1} \sqrt{\{x\}}, \{\cdot\}$ denotes fractional part function is

- (1) an even function
- (2) a periodic function
- (3) an odd function
- (4) neither even nor odd

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Q18 - Multiple Correct

The period of $f(x) = \sin \frac{\pi}{4}[x] + \cos \frac{\pi x}{2} + \cos \frac{\pi}{3}[x]$, where, $[\cdot]$ denotes greatest integer function, is

- (1) 8
- (2) 12
- (3) 24
- (4) None of these

Q19 - Multiple Correct

The period of function $f(x) = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right] - 3x + 15$ is

- (1) $\frac{1}{3}$
- (2) $\frac{2}{3}$
- (3) 1
- (4) non-periodic

Q20 - Multiple Correct

The fundamental period of

$f(x) = x + a - [x + b] + \sin \pi x + \cos 2\pi x + \sin 3\pi x + \cos 4\pi x + \cdots + \sin(2n - 1)\pi x + \cos 2n\pi x$ for every $a, b \in R$ (where, $[\cdot]$ denotes greatest integer function), is

- (1) 2
- (2) 4
- (3) 1
- (4) 3

Q21 - Multiple Correct

If $f : R \rightarrow [-1, 1]$, where $f(x) = \sin \frac{\pi}{2}[x]$ (where, $[\cdot]$ denotes greatest integer function), then

- (1) $f(x)$ is even
- (2) $f(x)$ is odd

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(3) $f(x)$ is periodic

(4) None of these

Q22 - Multiple Correct

Identify the statement(s) which is/are incorrect?

(1) The function $f(x) = \cos(\cos^{-1} x)$ is neither odd nor even

(2) The fundamental period of $f(x) = \cos(\sin x) + \cos(\cos x)$ is π

(3) The range of the function $f(x) = \cos(3 \sin x)$ is $[-1, 1]$

(4) None of the above

Q23 - Multiple Correct

If $f(x) = \frac{\sin \pi[x]}{\{x\}}$, then $f(x)$ is

(1) periodic with fundamental period 1

(2) even

(3) range is singleton

(4) None of these

Q24 - Multiple Correct

Let $f(x) = |x - 1| + |x - 2| + |x - 3| + |x - 4|$, then

(1) the least value of $f(x)$ is 4

(2) the least value is not attained at a unique point

(3) the number of integral solution of $f(x) = 4$ is 2

(4) the least value of $f(x)$ is 2

Q25 - Multiple Correct

Let $f(x) = \sec^{-1}[1 + \cos^2 x]$, where $[]$ denotes the greatest integer function, then

(1) the domain of f is R

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(2) the domain of f is $[1, 2]$ (3) the range of f is $[1, 2]$ (4) the range of f is $\{\sec^{-1} 1, \sec^{-1} 2\}$

Q26 - Paragraph 1

Passage I (For Question 26, 27)

If $f(x) = \frac{x+a}{bx^2+cx+2}$, where $a, b, c \in R$, $f(-1) = 0$ and $y = 1$ is an asymptote of $y = f(x)$, $y = f^{-1}(x)$ is inverse function of $f(x)$

 $f(x)$ is equal to

(1) $\frac{x+1}{x+2}$

(2) $\frac{x+1}{x^2+x+2}$

(3) $\frac{x+1}{2-x}$

(4) $\frac{x-1}{2x+1}$

Q27 - Paragraph 1

If $f^{-1} : R - \{1\} \rightarrow R$, then $f^{-1}(x)$ is

(1) one-one and onto

(2) one-one and into

(3) many-one and into

(4) many-one and onto

Q28 - Paragraph 2

Passage II (For Question 28, 29)

Let $f(x) = x^2 - 2x - 1, \forall x \in R$ and $f : (-\infty, a] \rightarrow [b, \infty)$, where a is the largest real number for which $f(x)$ is bijective.

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The value of $(a + b)$ is

- (1) -2
(2) -1

(3) 0

(4) 1

Q29 - Paragraph 2

Let $f : R \rightarrow R, g(x) = f(x) + 3x - 1$, then the least value of function $y = g(|x|)$ is

- (1) $-9/4$
(2) $-5/4$

(3) -2

(4) -1

Q30 - Paragraph 3

Passage III (For Question 30, 31)

Given a function $f_1(x) = e^{\tan\{\frac{x}{4}\}} + \cos \pi \left(\frac{(1-2[x])}{2} \right) + \sin \left(\frac{\pi[x]}{2} \right)$, whose fundamental period is p , (where $[\cdot]$ and $\{\cdot\}$ represents greatest integer function and fractional part functions respectively) and

$y = \sqrt{2p + \frac{p}{2}[x] - [x]^2}$. The domain of y is $[q, r)$ one another function $f_2(x) = \begin{cases} 2+x, & x \geq 0 \\ 2-x, & x < 0 \end{cases}$

The period p of $f_1(x)$ is

- (1) an irrational number
(2) a prime number
(3) a composite number
(4) neither a prime nor a composite number

Q31 - Paragraph 3

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The value of $(r - q - 1)$

(1) 6

(2) 7

(3) 8

(4) 9

Q32 - Paragraph 4

Passage IV (For Question 32, 33)

A tower has the shape as : A truncated right circular cone (one with radii $2R$ (the lower base) and R (the upper base) and the height R bears a right circular cylinder whose radius is R , the height being $2R$. Finally, a semisphere of radius R is mounted on the cylinder, suppose that the cross-sectional area S of the tower is given by $f(x)$, where x is the distance of the cross-section from the lower base of the cone.

The domain of $f(x)$ is

(1) $(-\infty, \infty)$

(2) $[0, 4R]$

(3) $[0, R]$

(4) $[R, 4R]$

Q33 - Paragraph 4

For $0 \leq x \leq R$, the function $f(x)$ is given by

(1) $\pi(2R - x)^2$

(2) $\pi(R - x)^2$

(3) πR^2

(4) $4\pi x^2$

Q34 - Paragraph 5

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Passage V (For Question 34, 35)

If $\{f(x)\}^2 \times f\left(\frac{1-x}{1+x}\right) = 64x, x \neq 0, 1$, then

$f(x)$ is equal to

(1) $4x^{2/3} \left(\frac{1+x}{1-x}\right)^{\frac{1}{3}}$

(2) $x^{1/3} \left(\frac{1-x}{1+x}\right)^{\frac{1}{3}}$

(3) $x^{2/3} \left(\frac{1-x}{1+x}\right)^{\frac{2}{3}}$

(4) None of these

Q35 - Paragraph 5

The domain of $f(x)$ is

(1) $[0, \infty)$

(2) $R - \{1\}$

(3) $(-\infty, \infty)$

(4) None of these

Q36 - Integer Type

$f(x) = [x] + \sum_{r=1}^{2015} \frac{\{x+r\}}{2015}$, then the value of $f(3)$ is

Q37 - Integer Type

The period of $f(x)$ satisfying $f(x+a) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2}$ is (λa) . Then, the value of λ is

Q38 - Integer Type

The period of $f(x)$ satisfying $f(x-1) + f(x+3) = f(x+1) + f(x+5)$, is

Q39 - Integer Type

Questions with Answer Keys

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The number of pairs (x, y) of real numbers with $0 < x < \frac{\pi}{2}$ such that $\frac{(\sin x)^{2y}}{(\cos x)^{y^2/2}} + \frac{(\cos x)^{2y}}{(\sin x)^{y^2/2}} = \sin 2x$, is

Q40 - Integer Type

Let $f(x)$ be a polynomial of degree 4 such that $f([i]) = \frac{1}{1+[i]}$ for $0 \leq i \leq 4$ (where $[x]$ denotes greatest integer function), then $f(6)$ is (use $f(5) = 1/3$)

Q41 - Integer Type

If p is a positive integer and f be a function defined for positive numbers and attains only positive values such that $f[xf(y)] = x^p y^4$, then p is

Q42 - Integer Type

Let $f : \left[-\frac{\pi}{3}, \frac{\pi}{6}\right] \rightarrow B$. Defined by $f(x) = 2 \cos^2 x + \sqrt{3} \sin 2x + 1$, where $B \in [a, b]$ such that $f^{-1}(x)$ exists, then $(a + b)$ is

Q43 - Integer Type

Let R be the region in the first quadrant bounded by the X and Y -axes and the graph of $f(x) = \frac{9}{25}x + b$ and $y = f^{-1}(x)$. If the area of R is 49, where the value of b is $\frac{\lambda}{\mu}$, then $|\lambda - 5\mu|$ is

Q44 - Integer Type

The function $f : [2, \infty) \rightarrow Y$ defined by $f(x) = x^2 - 4x + 5$ is both one-one and onto, if $Y \in [a, \infty)$, then the value of a is

Q45 - Integer Type

If $f(x) = \left(\frac{4a-7}{3}\right)x^3 + (a-3)x^2 + x + 5$ is one-one function, where $a \in [\lambda, \mu]$, then the value of $|\lambda - \mu|$ is

Q46 - Integer Type

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If $n(A) = 4, n(B) = 5$, number of functions from A to B such that range contains exactly 3 elements is k , then the value of $\frac{k}{60}$ is

Q47 - Integer Type

If $f(x)$ be a polynomial of degree 4 with leading coefficient 1 satisfying $f(1) = 10, f(2) = 20, f(3) = 30$, where $f(12) + f(-8) = (3968)\lambda$, then the value of λ is

Q48 - Integer Type

If $f(x) = \begin{cases} 1 + x^2, & x \leq 1 \\ x + 1, & 1 < x \leq 2 \end{cases}$ and $g(x) = 1 - x, -2 \leq x \leq 1$, thus

$(f \circ g)(x) = \begin{cases} 2 - 2x + x^2, & a \leq x \leq b \\ 2 - x, & c \leq x < a \end{cases}$. Then, the value $(a + b + c)$ is

Q49 - Integer Type

Let $f(x) = \begin{cases} 1 + x, & 0 \leq x \leq 2 \\ 3 - x, & 2 < x \leq 3 \end{cases}$. Thus, $(f \circ f)(x) = \begin{cases} 2 + x, & a \leq x \leq b \\ 2 - x, & b < x \leq c \\ 4 - x, & c < x \leq d \end{cases}$, then the value of $(a + b + c + d)$ is

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Answer Key					
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Q1 (2)	Q2 (3)	Q3 (2)	Q4 (1)		
// mathongo	// mathongo	// mathongo	// mathongo	// mathongo	// mathongo
Q5 (3)	Q6 (2)	Q7 (4)	Q8 (4)		
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Q9 (4)	Q10 (2)	Q11 (4)	Q12 (2)		
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Q13 (3)	Q14 (1)	Q15 (4)	Q16 (2)		
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Q17 (1, 2)	Q18 (3)	Q19 (1, 2, 3)	Q20 (1)		
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Q21 (3)	Q22 (1, 2, 3)	Q23 (2, 3)	Q24 (1, 2, 3)		
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Q25 (1, 4)	Q26 (1)	Q27 (2)	Q28 (2)		
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Q29 (3)	Q30 (3)	Q31 (1)	Q32 (2)		
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Q33 (1)	Q34 (1)	Q35 (2)	Q36 (3)		
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Q37 (2)	Q38 (8)	Q39 (1)	Q40 (1)		
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Q41 (2)	Q42 (4)	Q43 (3)	Q44 (1)		
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Q45 (6)	Q46 (6)	Q47 (5)	Q48 (0)		
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Q49 (6)					
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